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XDR Profile

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ABSTRACT
The focus of eHealth DSI is on enabling a health professional at a point of care to obtain medical information about a foreign patient. In Specifications, the use cases are generalized into "Provisioning of Data" interaction pattern that defines the communication flow between a health professional and medical data managing services (e.g. a Personal Health Record) in a foreign country.

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V1.1	17/12/2013	Integration of Review Comments	Fraunhofer FOKUS	P. Loubjerg, M. Sinke

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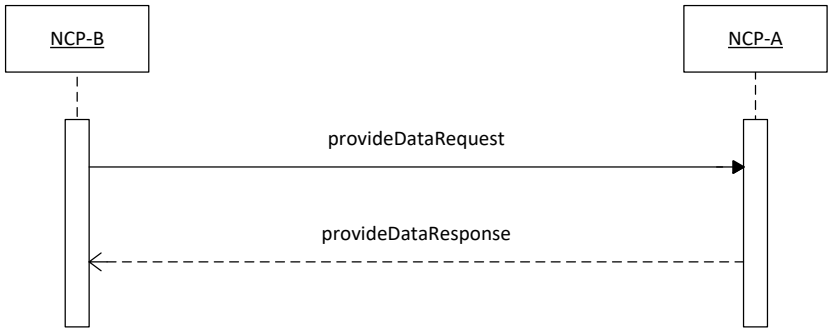
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1 Introduction

The focus of eHealth DSI is on enabling a health professional at a point of care to obtain medical information about a foreign patient. In [Interoperability Specifications], the use cases are generalized into "Provisioning of Data" interaction pattern that defines the communication flow between a health professional and medical data managing services (e.g. a Personal Health Record) in a foreign country.

1.1 eHealth DSI Country-B Document Provisioning

The "Provisioning of Data" interaction pattern is defined along the end-to-end communication between a health professional and a medical data managing service. Part of this logical end-to-end interaction is the communication between the countries' National Contact Points (NCPs) which facilitate the cross-border transferal of medical data. The respective NCP-to-NCP communication pattern is realized by the eHealth DSI "Provide Data" Service which implements the necessary message exchanges within the eHealth DSI domain (see figure 1).



The "Provide Data" service is always initiated by NCP-B as the document source with NCP-A acting as the document recipient. The service contract consists of two messages: *provideDataRequest* and *provideDataResponse*:

1. In order to transmit data of an identified patient to country A, the document source MUST first send a *provideDataRequest* message to the document recipient. This message carries the provided documents and (in case documents are provided in multiple formats) information about the relationship among these documents.
2. The document recipient MUST respond to a *provideDataRequest* message by sending back a *provideDataResponse* message to the document source. This message either acknowledges the receipt of the data or rejects the data (e.g. in cases where country-A does not implement means for processing or forwarding the provided data).

1.2 IHE Cross-Enterprise Document Reliable Interchange (XDR)

The eHealth DSI specification [Interoperability Specifications] defines the logical interfaces, the behaviour, and the overall semantics of the ProvideData Service but does not mandate for a specific technical standard for implementing the ProvideData request and response messages. This document specifies a respective binding onto the IHE Cross-Enterprise Document Reliable Interchange (XDR) integration profile.

1.2.1 Profile Overview

In [IHE ITI TF-1] the functionality and semantics of the XDR integration profile is defined as:

Cross-Enterprise Document Reliable Interchange (XDR) provides document interchange using a reliable messaging system. […] Transfer is direct from source to consumer, no repository or registry actors are involved. […] XDR is document format agnostic […]
In the eHealth DSI context the messaging infrastructure and security framework of the NCP-to-NCP eHealth DSI domain provide the reliable messaging platform to XDR. In contrast to the typical data retrieve pattern as specified in [XCA Profile], this binding is specifying a data provision initiated in Country-B with a destination in Country-A. Therefore, NCP-B takes the role of the document source while NCP-A acts as a proxy and document dispatcher for the document recipient located at the Country-of-Affiliation (Country-A) of the patient.

1.2.2 eHealth DSI Profiling of the XDR Integration Profile

Within eHealth DSI the actors and transactions of the IHE XDR integration profile are mapped onto eHealth DSI services and messages as follows:

IHE XDR	**eHealth DSI Provide Data Service**
Actor: Document Source	NCP-B
Actor: Document Recipient	NCP-A
Transaction: Provide and Register Document Set (ITI-41)	ProvideData request and response messages

This eHealth DSI binding introduces minor extensions and restrictions on the IHE XDR actor and transaction definitions in order to properly cover the eHealth DSI use cases and to align with the eHealth DSI security framework:

- Additional error messages are defined that cover specific failure conditions of the eHealth DSI use cases
- The optionality of data fields is aligned to eHealth DSI privacy requirements

The application of security measures and the contents of the SOAP security header are specified normatively (see [Messaging Profile]).

1.3 Related Documents

The functional specification of the Provide Data Service is defined in [Interoperability Specifications] which is the normative foundation for all binding specifications. The technical binding defined in this document makes direct use of other bindings:

- [Audit Trail Profile] for a specification of the eHealth DSI common audit trail contents and structure.
- [Messaging Profile] for a specification of how SOAP and Web Service Security are to be used for implementing eHealth DSI services.
- [SAML Profile] for a specification of the HP Identity Assertion and the Treatment Relationship Conformation Assertion which are used for providing authentic information about the consumer of an eHealth DSI service and the recent care context.

The medical documents to be transmitted by the Provide Data service are specified through Semantic Signifiers (see [Interoperability Specifications]). Examples of such signifiers are:

- eDispensation for notifying the patient's country of affiliation in the dispensation of a medication for that patient.
- eDispensation Discard for notifying the patient's country of affiliation.

1.4 Conventions

The keywords MUST, SHOULD, MAY, SHOULD NOT and MUST NOT are used as defined in [RFC 2119].

eHealth DSI requirements are managed within a central requirements management tool and serialized into [eHealth DSI Requirements Catalogue]. References to requirements always use the identifiers as used with the central requirements. Newly defined requirements are handed over to the central requirements management. In case of doubt or conflict, [eHealth DSI Requirements Catalogue] MUST be considered to be the normative phrasing of a requirement.

1.5 Terms and Definitions

The country which holds information about a patient, where the patient can be univocally identified and where his data may be accessed is called "country-A" (country of affiliation) [eHealth DSI Glossary]. The country of treatment where cross-border health care is provided when the patient is seeking care abroad is called "country-B" (country of care) [eHealth DSI Glossary].
The term "Healthcare Professional (HP)" denotes a doctor of medicine, a nurse responsible for general care, a dental practitioner, a midwife or a pharmacist within the meaning of Directive 2005/36/EC, or another professional exercising activities in the healthcare sector which are restricted to a regulated profession as defined in Article 3(1)(a) of Directive 2005/36/EC, or a person considered to be a health professional according to the legislation of the country of treatment [eHealth DSI Glossary]. Health professionals are allowed to process medical patient data according to the legislation of the country of the health professional's residence. An "eHealth DSI Point of Care (PoC)" is a location where an eHealth DSI patient may seek healthcare services [eHealth DSI Glossary].
A "National Contact Point (NCP)" is an entity in each particular country to act as a bidirectional technical, organizational and legal interface between the existing national functions and infrastructures [eHealth DSI Glossary]. In this document the term "NCP" emphasizes the technical aspects of this interface and as such refers to a gateway which facilitates various aspects of cross-border data sharing (e.g. message forwarding, signature verification). From an architectural perspective NCPs denote the boundary between the eHealth DSI infrastructure and a country's existing national eHealth infrastructure (see [Interoperability Specifications] for details).

1.6 Namespaces

XML namespace prefixes are used in this document to stand for their respective namespaces as follows.

Prefix	Namespace
ehealth	urn:ehdsi:v1
soapenv	http://www.w3.org/2003/05/soap-envelope
wsse	http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-wssecurity-secext-1.0.xsd
saml	urn:oasis:names:tc:SAML:1.0:assertion
xacml	urn:oasis:names:tc:xacml:2.0:policy:schema:os
hl7v2	urn:hl7-org:v2
hl7v3	urn:hl7-org:v3
xs	http://www.w3.org/2001/XMLSchema
xsi	http://www.w3.org/2001/XMLSchema-instance
query	urn:oasis:names:tc:ebxml-regrep:xsd:query:3.0
rim	urn:oasis:names:tc:ebxml-regrep:xsd:rim:3.0
rs	urn:oasis:names:tc:ebxml-regrep:xsd:rs:3.0
lcm	urn:oasis:names:tc:ebxml-regrep:xsd:lcm:3.0

1.7 Status of this Binding

The binding as defined in this document is a normative binding. All eHealth DSI countries MUST implement this binding within their NCP.
Other eHealth DSI documents must refer to THIS binding as [XDR Profile]. Unless a version number is given, references to [XDR Profile] always refer to the most current version that is published on the eHealth DSI website.
Additional or alternative bindings implementing the eHealth DSI Provide Data Service MUST NOT be defined and implemented for the eHealth DSI.

2 eHealth DSI Provide Data Service XDR Binding

This binding to the *eHealth DSI Provide Data Service* is compliant with the IHE profile *Cross-Enterprise Reliable Exchange* XDR as normatively specified in [IHE ITI TF-1] and [IHE ITI TF-XDR].

2.1 ProvideData Request Message

The ProvideDataRequest message is issued by an HP in country-B for transmitting data of an identified patient to that patient's country of affiliation. Each transmission carries one or more eHealth DSI pivot coded documents and/or the source coded documents that encode the same information without semantic mapping (see [Interoperability Specifications] for details).

The ProvideDataRequest message implements the *IHE Provide And Register DocumentSet* transaction (ITI-41) request message as profiled in [IHE ITI TF-XDR] considering the following constraints:

- The document recipient SHOULD process the provided documents as a single IHE XDS submission set acc. to [IHE ITI TF-2].
- The document recipient SHOULD ignore groupings and MUST ignore all associations between documents and submission sets.
- The document recipient MUST NOT process any metadata assigned to the submission set, it MUST solely rely on the document metadata and contents.

For each document (either eHealth coded or source coded) the following set of metadata MUST be provided:

Slot Name	Binding	Slot Value
id	Attribute	Identifier of the document. This identifier MUST be the same for: <rim:ExtrinsicObject/@id> and <ihe:Document/@id>.
mimeType	Attribute	MUST be "text/xml"
objectType	Attribute	MUST be set according to section 4.3.1.2 of [IHE IT TF 3]
status	Attribute	MUST be "urn:oasis:names:tc:ebxml-regrep:StatusType:Approved"
creationTime	rim:Slot	MUST be given for XDR compatibility. MAY be ignored by the document recipient.
languageCode	rim:Slot	MUST be given for XDR compatibility. MAY be ignored by the document recipient.
sourcePatientId	rim:Slot	MUST be of the same value as patientId (see below)
healthcareFacilityTypeCode	classification	MUST be provided for XDR compatibility and correct addressing. Value MUST be set to ISO 3166-1 alpha-2 country code of the providing PN.
practiceSettingCode	classification	MUST be provided for XDR compatibility but MAY be ignored by the document recipient. Value MUST be set to "Not Used".

confidentialityCode	classification	MUST be provided for XDR compatibility but MAY be ignored by the document recipient. Value SHOULD be set to "N".
classCode	classification	See section 2.1.1 (eHDSI Discard Medication is using the code DISCARD-60593-1 as no LOINC code are currently available).
formatCode	classification	Format qualifier as defined in [eHDSI CDA Implementation Guides].
patientId	External identifier	Equals to the patient identifier that was provided by the *eHealth DSI Identification Service* (encoded as HL7 v3 II data type)
typeCode	classification	MUST contain the LOINC code + coding scheme 2.16.840.1.113883.6.1 + name of the transported document. Coding MUST be provided according to ITI TF-2a 3.18.4.1.2.3.4 Coding of Code/Code-Scheme. eDispensation: 60593-1 and name "eDispensation" Patient Consent: 57016-8 and name "Privacy Policy Acknowledgement Document"
uniqueId	External identifier	MUST refer to the OID of the document that is included within the <ihe:Document> element.

Other metadata than the ones listed above SHOULD NOT be provided by the service provider[1]. If given they MUST be ignored by the service consumer.

2.1.1 Semantic Signifiers

eHDSI introduces the concept of a "Semantic Signifier" as a holistic specification and usage guidance for document classes. A semantic signifier is an XML structure that encapsulates conceptual models for defining the semantics of a document class, formal specifications (e.g. as Schemas and Schematrons) for implementing and verifying respective data objects, and approved transformation scripts (e.g. for visualization).

In order to preserve compatibility with [IHE ITI TF-2] XCA and epSOS Pilot implementations, eHDSI decouples the semantic signifier structure from the IHE X* metadata by using the LOINC-coded document class code as an indirect reference to the respective semantic signifier:

- classCodes MUST be coded according to specification in [ITI TF-2a] section 3.18.4.1.2.3.4 Coding of Code/Code-Scheme.
- For classCodes the classification scheme 2.16.840.1.113883.6.1 with codeSystemName="LOINC" MUST be used.
- For eHDSI document classes (ePrescription, eDispensation, Patient Summary, Laboratory Results Report and OrCD) the LOINC codes as defined in [eHDSI CDA Implementation Guides]. MUST be used for the respective document classes. Exception for eDispendation Discard using the code DISCARD-60593-1 as no LOINC code are currently available.

eHealth DSI Semantic Signifiers MUST be univocally mapped on LOINC Codes in order to allow an NCP to obtain a document class Semantic Signifier from the document class coded that is coded within an IHE X* message.

2.1.2 Associations

An ebRIM association MUST be used for declaring the eHealth DSI pivot coded document as a transformation of the source coded document. As classification scheme

[1] Document linkage information (e.g. a reference to the ePrescription that is affected by the dispensation) MUST be included with the document (see [eHDSI CDA Implementation Guides]) and MUST NOT be part of the message metadata.

urn:uuid:abd807a3-4432-4053-87b4-fd82c643d1f3 MUST be used per IHE-XDR. Currently "eHealth DSI pivot" is defined as the only valid transformation:

```
<rim:Association id="id of the association"
    associationType="urn:ihe:iti:2007:AssociationType:XFRM"
    sourceObject="UUID of the source coded document"
    targetObject="UUID of the eHealth DSI pivot document"
    objectType="urn:oasis:names:tc:ebxml-regrep:ObjectType:RegistryObject:Classification">
    <rim:Classification id="id of the classification"
      classificationScheme="urn:uuid:abd807a3-4432-4053-87b4-fd82c643d1f3"
      classifiedObject="id of the association"
      objectType="urn:oasis:names:tc:ebxml-regrep:ObjectType:RegistryObject:Classification"
      nodeRepresentation="eHealth DSI pivot">
      <rim:Slot name="codingScheme">
        <rim:ValueList>
            <rim:Value>eHealth DSI translation types</rim:Value>
        </rim:ValueList>
      </rim:Slot>
      <rim:Name>
        <rim:LocalizedString value="Translation into eHealth DSI pivot format"/>
      </rim:Name>
    </rim:Classification>
</rim:Association>
```

2.2 Expected Actions

The eHealth DSI Provide Data Service document source shall respond to a ProvideDataRequest message with the ProvideDataResponse message containing a success indicator.

The eHealth DSI Provide Data Service document recipient MUST verify that the requesting service user has sufficient rights to submit the given kind of documents for the identified patient.

In case of an error that relates to the transmission of the request or the processing of the eHealth DSI security token, the *eHealth DSI Provide Data Service* document recipient MUST respond with a fault message according to [Messaging Profile].

2.2.1 Response Message (Full Success Scenario)

If the eHealth DSI Provide Data Service document recipient is able to decode the received message and to properly process/forward all transmitted documents it responds with an *ebXML Registry Response* with its status set to: "urn:oasis:names:tc:ebxml-regrep:ResponseStatusType:Success".

If the document recipient wants to respond with further information on the processing of the transmitted data or with a non-critical warning it SHOULD include an additional <RegistryErrorList> element. The severity MUST be set to: "urn:oasis:names:tc:ebxml-regrep:ErrorSeverityType:Warning":

```
<rs:RegistryResponse
        status="urn:oasis:names:tc:ebxml-regrep:ResponseStatusType:Success">
  <rs:RegistryErrorList>
    <rs:RegistryError
        severity="urn:oasis:names:tc:ebxml-regrep:ErrorSeverityType:Warning"
        errorCode="...."
        codeContext="Processing deferred" location="" />
  </rs:RegistryErrorList>
</rs:RegistryResponse>
```

The following warning messages and codes are defined:

Condition and Severity	Message	Code	Action to be taken
Documents were received but not processed	Processing deferred	2201	None

2.2.2 Response Message (Failure or Partial Failure Scenario)

If the *eHealth DSI Provide Data Service* document recipient is able to decode the received message but the processing/forwarding of one or more documents failed, it responds with an *ebXML Registry Response* that contains a respective status indicator (see below). The response **MUST** contain a RegistryErrorList element that indicates the failure condition.
If none of the documents was processed successfully, the response status **MUST** be set to "urn:oasis:names:tc:ebxml-regrep:ResponseStatusType:Failure". If at least one document was processed successfully, the response status **MUST** be set to "urn:ihe:iti:2007:ResponseStatusType:PartialSuccess".

Note that multiple registry error messages **MAY** be included within a single <rs:RegistryErrorList> element.

Additional information to describe the reason **MUST** be included in the acknowledgementDetail of the response.

AcknowledgementDetail.Code	The value **MUST** be a code listed in the tables of the Appendix 6.4 of [NCPeH Components Specifications].https://webgate.ec.europa.eu/fpfis/wikis/x/n_PzN
AcknowledgementDetail.Text	The value is a text description of the error code
AcknowledgementDetail.Location	A failure location **MUST** be provided if the error does not apply to all provided documents. It **MUST NOT** be given if the error applies to all provided documents

For more information see Exception_Handling in MyHealth@EU

3 Security Considerations

3.1 Application of eHealth DSI Security Services

Implementations of this binding MUST consider the eHealth DSI standard security safeguards in order to preserve basic protection for data confidentiality, data integrity and patient privacy:

- All messages MUST make use of the Web Service Security Standard and the WS Addressing Standard as defined in [Messaging Profile].
- Each request message MUST contain claims on the identity of the requestor and on the care context as defined in [SAML Profile] and [Messaging Profile].
- The document recipient MUST verify the integrity and authenticity of any assertions and other security token communicated alongside for each incoming request.
- Before accepting provided data the document recipient MUST verify the existence of a valid consent and assess all applicable security policies.

Before accepting provided data the NCP-A MUST verify that it is capable of forwarding the received data to its final destination within the CoA.

3.2 Security Audit Considerations

The service consumer MUST write an audit trail entry according to the *HP Assurance Audit Schema* as defined in [Audit Trail Profile]. The service provider MUST write an

audit trail entry according to the *Patient Privacy Audit Schema* as defined in [Audit Trail Profile].
The following table defines which categories MUST be filled (R), which MAY be filled (O) and which categories MUST NOT be used (X).

eHealth DSI Instance	Opt.[2]	Description
Event	R	Audited event
Requesting Point of Care	R / X	HPCO that issued the original request. This category MUST be filled by the service consumer. It MUST NOT be provided by the service provider.
Human Requestor	R	HP that triggered the request
Source Gateway	R	Service consumer node address at the country of Care
Target Gateway	R	Service provider node address at the country of the patient's affiliation
Audit Source	R	Legal entity that ensures the uniqueness of the identifiers that are used to identify active participants
Patient	R	Patient
Event Target	R	References to the provided documents (see below)
Error Message	O	Only used in case that the request handling was not completed successfully or for anomaly detection

For the Event Target Category the following fields MUST be provided:

Field Name	Opt.	Value Constraints
ParticipantObjectTypeCode	R	MUST be "2" (System Object)
ParticipantObjectTypeCodeRole	R	MUST be "4" (Resource)
ParticipantObjectIDTypeCode	R	MUST be "12" (URI)
ParticipantObjectID	R	MUST be string-encoded UIDs of the provided documents

4 References

4.1 Normative References

[IHE ITI TF-1] IHE International*: IT Infrastructure Technical Framework Volume 1 – Integration Profiles*. Revision 19, June 2022. https://profiles.ihe.net/ITI/TF/Volume1/index.html

[IHE ITI TF-2] IHE International: IHE IT Infrastructure (ITI) Technical Framework. Volume 2: Transactions. Revision 19, June 2022. https://profiles.ihe.net/ITI/TF/Volume2/index.html

[IHE ITI TF-XDR] IHE International*: IT Infrastructure Technical Framework Provide and Register Document Set-b*. Revision 19, June 2022. https://profiles.ihe.net/ITI/TF/Volume2/ITI-41.html

[IHE ITI TF-3] IHE International: IHE IT Infrastructure (ITI) Technical Framework. Volume 3 – *Integration Profiles*. Revision 19, June 2022. https://profiles.ihe.net/ITI/TF/Volume3/index.html

[RFC 2119] Bradner, S.: Key words for use in RFCs to Indicate Requirement Levels. Harvard University, Boston, Massachusetts, 1997.

[2] Categories MUST be filled (**R**), which MAY be filled (**O**) and which categories MUST NOT be used (**X**).